Angles in Triangles and Quadrilaterals

Name _____ Date _____ No calculator Version 1

Scan to practise this topic online

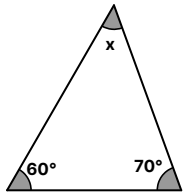
KEY WORDS triangle *(tapatoru)* · quadrilateral *(tapawhā)* · angle sum · right angle *(koki hāngai)*

A Angles in a triangle ●○○

Remember: The three angles inside any triangle add to 180°.

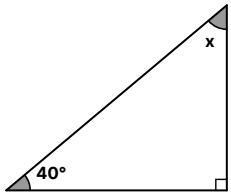
1. Find the size of the angle marked x.

a)



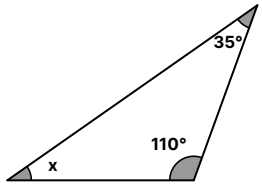
_____

b)



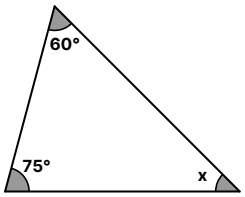
_____

c)



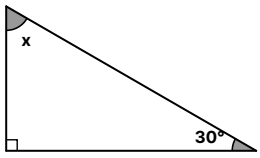
_____

d)



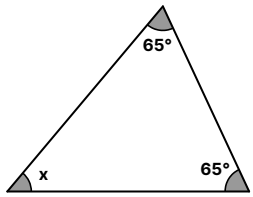
_____

e)



_____

f)



_____

2. Answer each question.

a) Two angles of a triangle are 35° and 80°. Find the third angle. _____

b) Two angles of a triangle are 90° and 47°. Find the third angle. _____

c) Two angles of a triangle are 118° and 26°. Find the third angle. _____

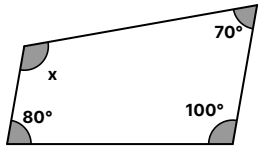
© 2026 Cell Tuition. You may print and photocopy this worksheet for your own classes and children. It may not be changed, sold or posted on other websites.

B Angles in a quadrilateral ●●○

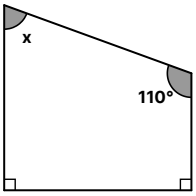
Remember: The four angles inside any quadrilateral add to 360°.

3. Find the size of the angle marked x.

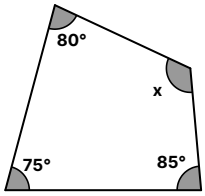
a)



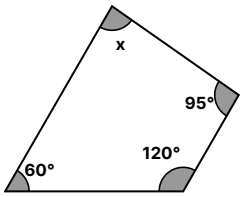
b)



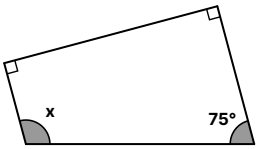
c)



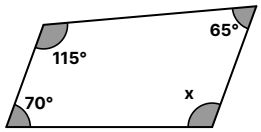
d)



e)



f)



4. Answer each question.

a) Three angles of a quadrilateral are 90°, 85° and 110°. Find the fourth angle. _____

b) Three angles of a quadrilateral are 72°, 108° and 64°. Find the fourth angle. _____

c) Three angles of a quadrilateral are 90°, 90° and 90°. Find the fourth angle. _____

C Could it be drawn? ●●●

Remember: Add the angles. A triangle needs a total of exactly 180°, and a quadrilateral needs exactly 360°.

5. Could a triangle have these three angles? Show your working.

a) 60°, 60°, 60°

b) 90°, 50°, 50°

c) 100°, 45°, 35°

d) 70°, 70°, 30°

6. Could a quadrilateral have these four angles? Show your working.

a) 90°, 90°, 90°, 90°

b) 100°, 80°, 110°, 60°

c) 120°, 60°, 120°, 60°

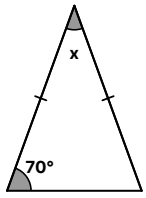
d) 95°, 85°, 100°, 80°

★ Extension

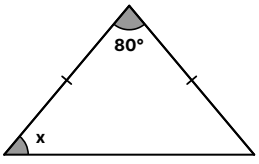
7. One angle of a right-angled triangle is 35°. Find the size of its third angle.

8. The small lines mark the equal sides. Find the size of the angle marked x. Give a reason.

a)



b)



© 2026 Cell Tuition. You may print and photocopy this worksheet for your own classes and children. It may not be changed, sold or posted on other websites.

Angles in Triangles and Quadrilaterals

Name _____ Date _____ No calculator Version 2

Scan to practise this topic online

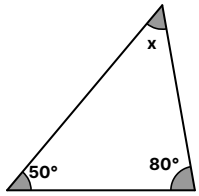
KEY WORDS triangle *(tapatoru)* · quadrilateral *(tapawhā)* · angle sum · right angle *(koki hāngai)*

A Angles in a triangle ●○○

Remember: The three angles inside any triangle add to 180°.

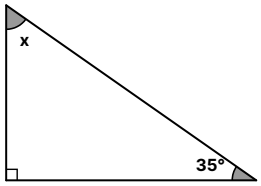
1. Find the size of the angle marked x.

a)



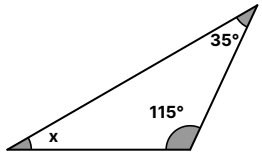
_____

b)



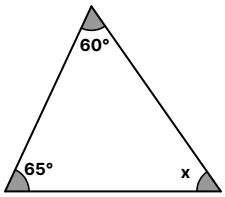
_____

c)



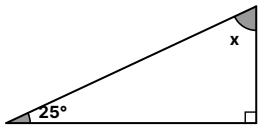
_____

d)



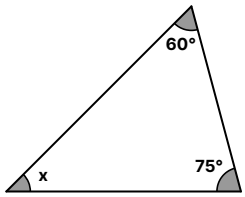
_____

e)



_____

f)



_____

2. Answer each question.

a) Two angles of a triangle are 45° and 70°. Find the third angle. _____

b) Two angles of a triangle are 90° and 53°. Find the third angle. _____

c) Two angles of a triangle are 124° and 19°. Find the third angle. _____

© 2026 Cell Tuition. You may print and photocopy this worksheet for your own classes and children. It may not be changed, sold or posted on other websites.

B Angles in a quadrilateral ●●○

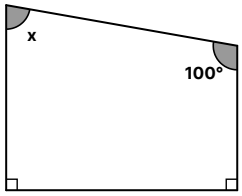
Remember: The four angles inside any quadrilateral add to 360°.

3. Find the size of the angle marked x.

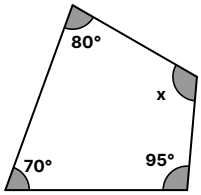
a)



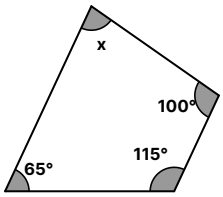
b)



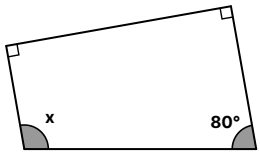
c)



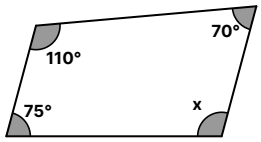
d)



e)



f)



4. Answer each question.

a) Three angles of a quadrilateral are 90°, 95° and 105°. Find the fourth angle. _____

b) Three angles of a quadrilateral are 68°, 112° and 74°. Find the fourth angle. _____

c) Three angles of a quadrilateral are 90°, 90° and 90°. Find the fourth angle. _____

C Could it be drawn? ●●●

Remember: Add the angles. A triangle needs a total of exactly 180°, and a quadrilateral needs exactly 360°.

5. Could a triangle have these three angles? Show your working.

a) 45°, 45°, 90°

b) 80°, 60°, 50°

c) 110°, 40°, 30°

d) 75°, 75°, 40°

6. Could a quadrilateral have these four angles? Show your working.

a) 90°, 90°, 100°, 80°

b) 110°, 70°, 120°, 50°

c) 130°, 50°, 130°, 50°

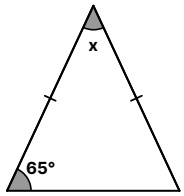
d) 95°, 95°, 80°, 80°

★ Extension

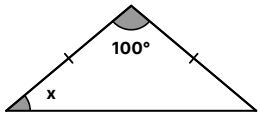
7. One angle of a right-angled triangle is 28°. Find the size of its third angle.

8. The small lines mark the equal sides. Find the size of the angle marked x. Give a reason.

a)



b)



© 2026 Cell Tuition. You may print and photocopy this worksheet for your own classes and children. It may not be changed, sold or posted on other websites.